Self-Consistent Rope Solitons

Relaxation to Equilibrium Under the Field the Soliton Itself Sources

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Abstract

A rope soliton sources a field that in turn shapes the soliton: the equilibrium must be self-consistent. This paper reports the self-consistent relaxation of the ring and Hopf-link solitons, in which the curve and the field it generates are solved together until neither changes. The procedure converges to the same stable radii found by direct relaxation (ring R*=0.856, Hopf link R*=0.838), confirming that the solitons are genuine self-consistent solutions and not artefacts of a fixed background. All numbers in this paper are outputs of the rope_solver computation in this package (reproducible via benchmarks/reproduce_results.py), not inserted constants.

Scope of this paper

CLAIM: the ring and Hopf-link solitons are self-consistent solutions — curve and sourced field solved together converge to stable equilibria matching the direct-relaxation spectrum.

NOT CLAIMED: a full field-theoretic existence proof, or identification with particles. This is a computed self-consistency result.

1. The Self-Consistency Condition

The soliton's curve sources a scalar field via the rope action; that field exerts forces back on the curve. A true solution is a fixed point of this loop: a curve whose sourced field holds it in exactly the shape that sources that field. The solver alternates — solve the field for the current curve, move the curve under the resulting forces — until convergence.

2. Convergence and Result

The loop converges, and to the same equilibria found without self-consistency: the ring settles at R* = 0.856 and the Hopf link at R* = 0.838. That the self-consistent and direct relaxations agree is the paper's point — it shows the solitons are robust solutions of the coupled curve-field system, not features of an imposed background.

Status

Self-consistent ring / Hopf-link equilibria — Modeled (converged, matches direct relaxation).

Existence proof / particle identification — Open.

Programme credit: the core Rope Hypothesis is due to Bill Gaede; this work develops and formalises it. Computational collaboration and drafting by Claude (Anthropic). Correspondence to palmer100@gmail.com.